

Ques

to implement

clk div by 3

→ code is completing in 3 cycles

But output is ~~completing~~ slowing down.

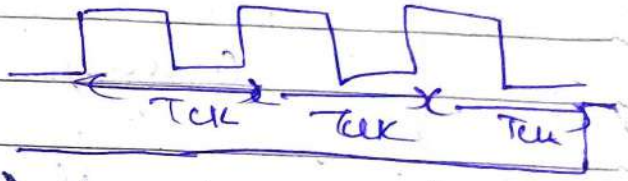
MOD-3 counter
Hota hai

$$\frac{N}{3} \leq 2$$

∴ 2 bit counter needed

00
01
10

fn(3)



module clk-div-3(
input clk, rst, output clkout)

internal reg

reg [1:0] count;

always @ (posedge clk) begin

if (rst)

count <= 0;

else if (count < 2)

count <= count + 1'b1;

else

count <= 0

end

assign clkout = count[1];

endmodule

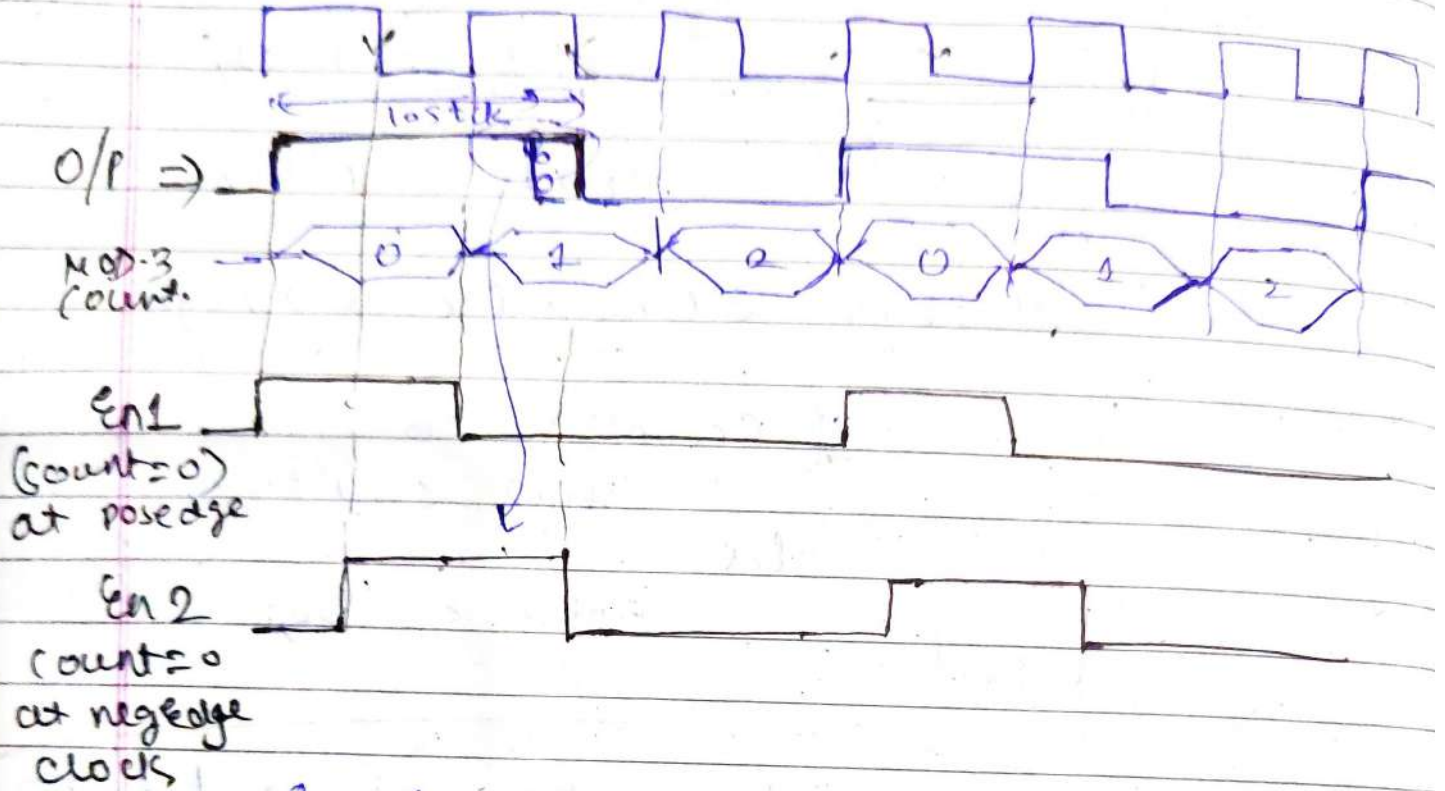
✓ MOD-3 counter

$f = \frac{1}{3}$
 $T = 3 \text{ ticks}$

$\frac{50 \times 3 \text{ ticks}}{100 \text{ (T)}} = 1.5 \text{ ticks} = T_{on}$

$DT = \frac{T_{on}}{T_{on} + T_{off}}$

$f/3$ with 50% DT $\rightarrow$ 50% half on, 50% half off



input clk, rst, ~~en1~~
 output fby3_50dt;

reg [1:0] counter;
 reg en1, en2;

always @ (posedge clk) begin

if (rst)
 counter <= 0;
 en1 <= 0;
 en2 <= 0;

end
 else

if (counter < 2)
 counter <= counter + 1'b1;

else
 counter <= 0;

end

MOD-3

```

always @ (posedge clk) begin
    if (counter == 0)
        enb1 <= 1'b1;
    else
        enb1 <= 1'b0;
end

```

```

always @ (negedge clk) begin
    if (counter == 0)
        enb2 <= 1'b1;
    else
        enb2 <= 1'b0;
end

```

```

assign fby350dt = enb1 | enb2;

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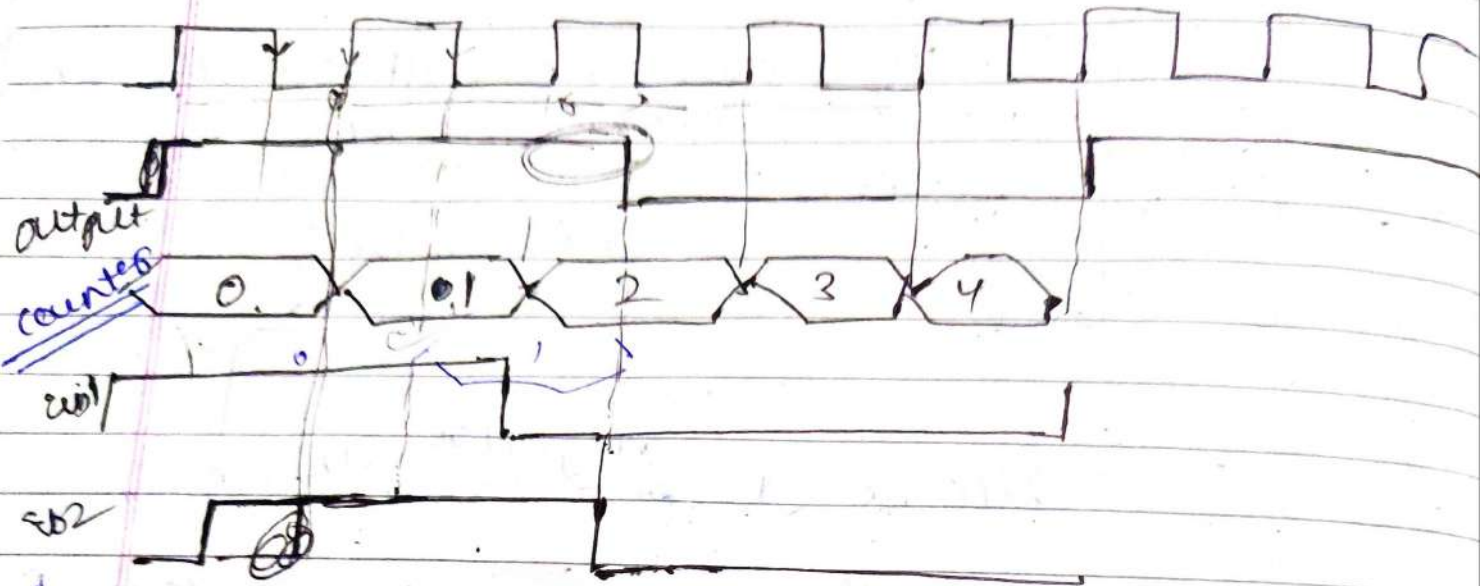
```

endmodule

```

$\frac{1}{5} \text{ base} \rightarrow T_0 = 5 T_{clk}$
 $DT = \frac{50}{100} = \frac{T_{on} T_{off}}{T_{on} + T_{off}}$
 $T_{on} = \frac{1}{2} (5 T_{clk}) = 2.5 T_{clk}$

f/5 with 50% DT



begin when count = 0
 & count = 1
 at negedge clk.

module (input clk, rst, output reg clkout)

reg [2:0] counter;
 reg en1, en2;

always @(posedge clk) begin

if (rst) begin

counter <= 0

en1 <= 0

en2 <= 0

end else if (counter < 4)

counter <= counter + 1;

else

counter <= 0

end

MOD-5

always @ (posedge clk) begin
 if (counter == 0 || counter == 1)
 en1 <= 1'b1;
 else
 en1 <= 1'b0;
 end

always @ (posedge clk) begin
 if (counter == 0 || counter == 1)
 en2 <= 1'b1;
 else
 en2 <= 1'b0;
 end

assign clkdiv_sout = en1 | en2;

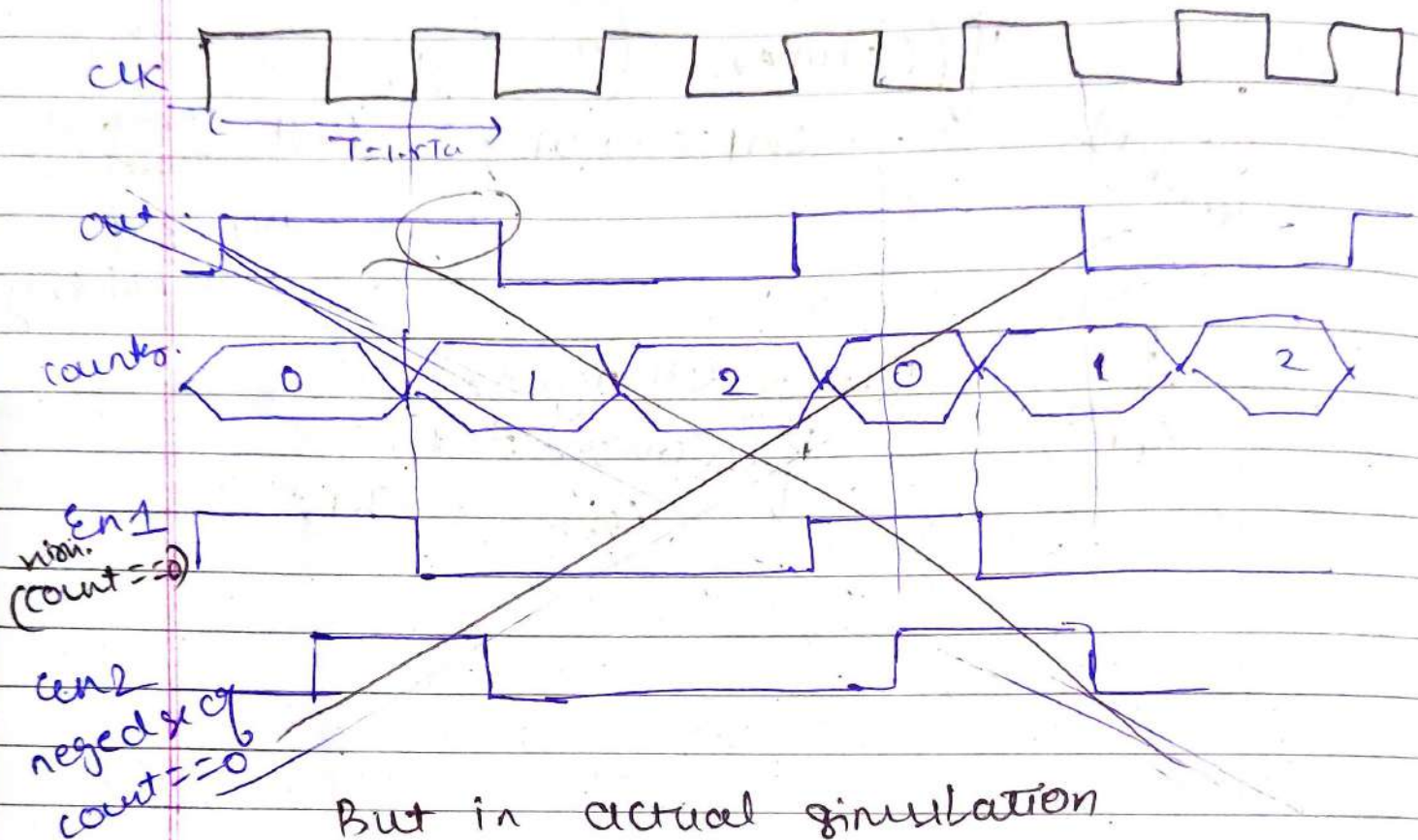
endmodule

$f_{clk}/105$ divider.



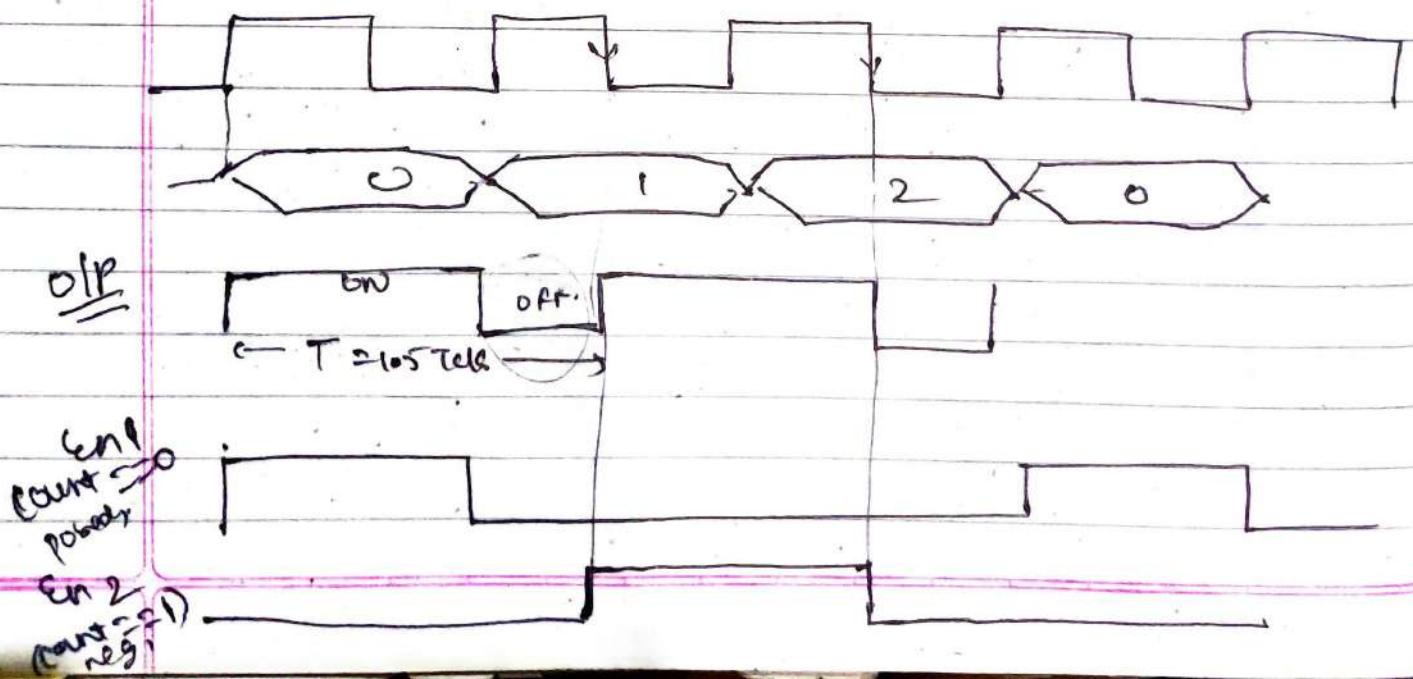
$T = 105 T_{clk}$
Total period

counts 2 bit



But in actual simulation

en1 get High at count == 1



So in ~~the~~ simulation tool.

always (@posedge clk)

for
enable1

if (count == 2)

en1 <= 'b1;

then it sets to high when at count = 0

BeCOZ of
Non Blocking

for
en2

always @(negedge clk)

if (count == 1)

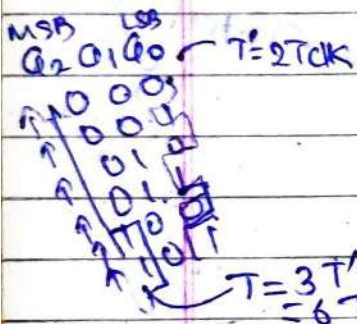
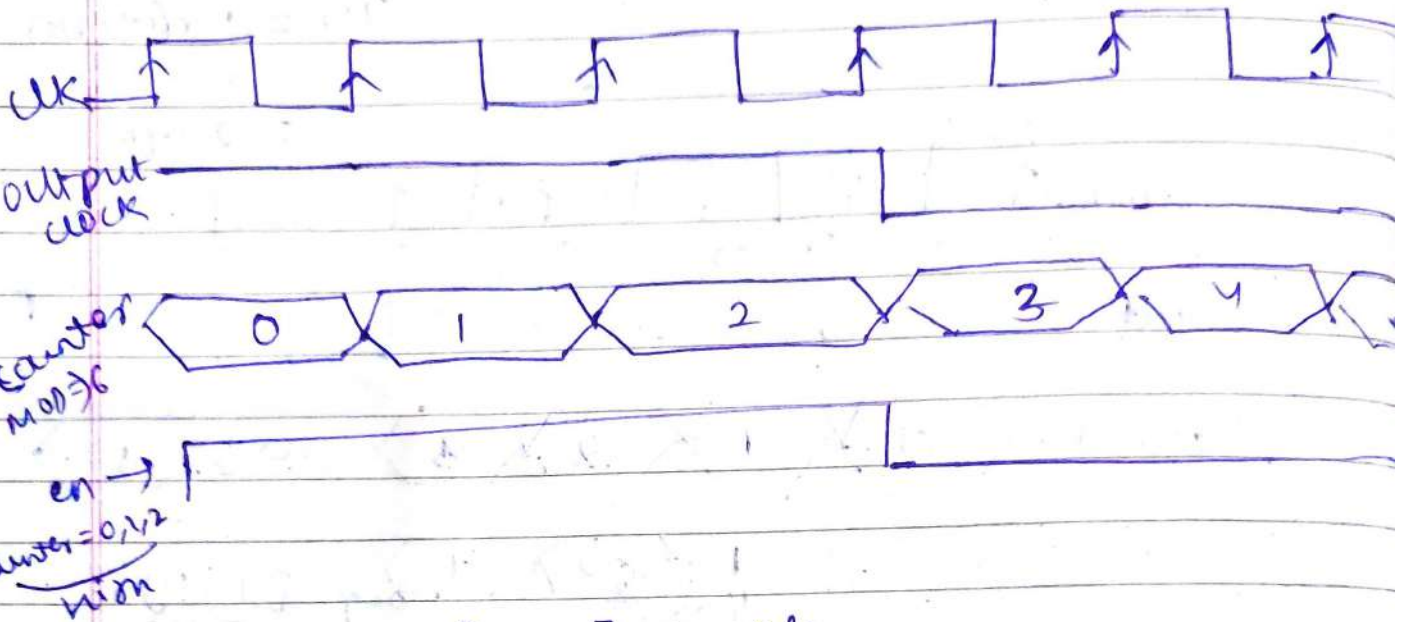
en2 <= 'b1;

$$T = 6 T_{clk}$$

$$T_{on} \Rightarrow \frac{1 \times 3 T_{clk}}{2} = 3 T_{clk}$$

MOD-6
6 < 2³

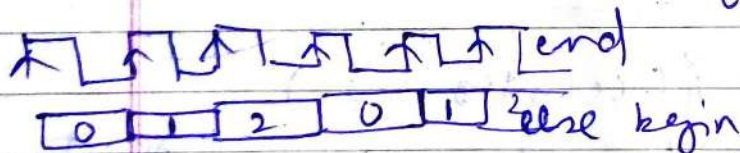
reg/6 with 50% DT



reg [2:0] counter;

always @(posedge clk) begin
if (rst) begin

counter <= 0;
output_clk <= 0;



if (counter == 2) begin
output_clk <= ~output_clk;
counter <= 0;
end else begin
counter <= counter + 1;
end

end end

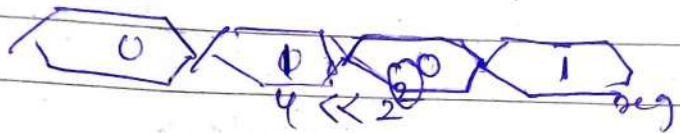
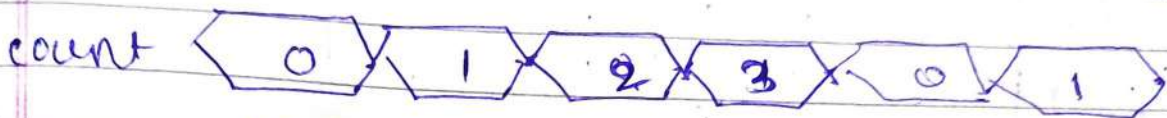
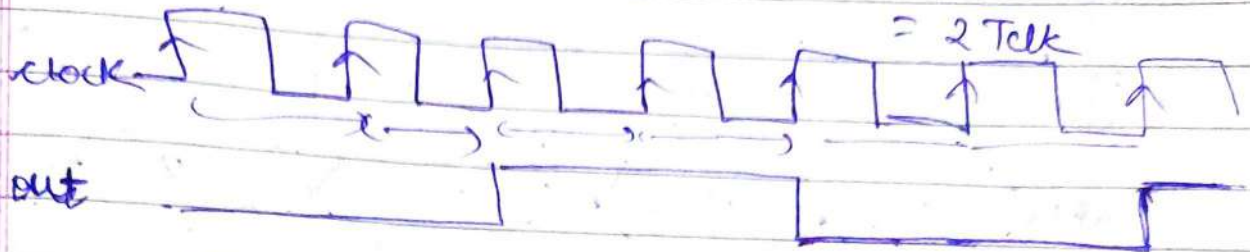
OR if you take MOD-6 then en → high for count=0, 1, 2
But in actual RTL
[count == 0, 1, 5] keep

6/4

50'1.DT

$T = 4T_{clk}$

$$T_{on} = \frac{1}{2} \times (T_{clk})$$



$[1:0]$ Count
2 bit
Total 4 counts
0
1
2
3

always @ () begin

if (xst)

else if (counter == 1)

counter < 20 ✓ reset

out = ~ outp - ck

✓ toggle

else

count = count + 1;

end

or

counter == 1 || counter == 3

output - ck = ~ output - ck

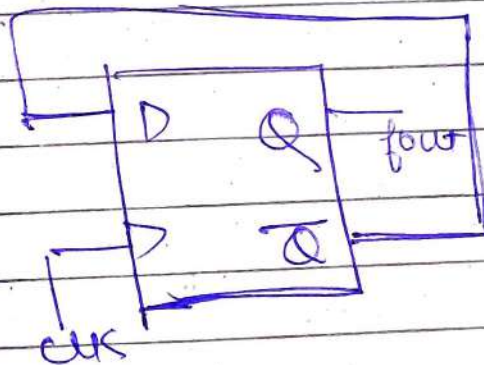
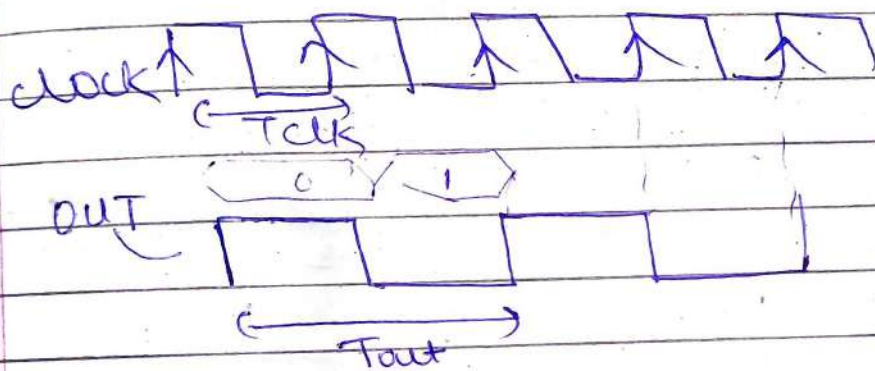
then we don't reset the counter

Q 50% Duty cycle frequency divide by 2

$T_{on} = 50\% \cdot T \rightarrow$
 $T_{on} = \frac{T}{2} \rightarrow f_{out} = \frac{f_{clk}}{2}$

$T_{off} = \frac{T}{2}$

$T = 2T_{clk}$



25% DT with $\frac{f_{clk}}{2}$

$$T_{on} = \frac{25}{100} T$$

$$f_{out} = \frac{f_{clk}}{2}$$

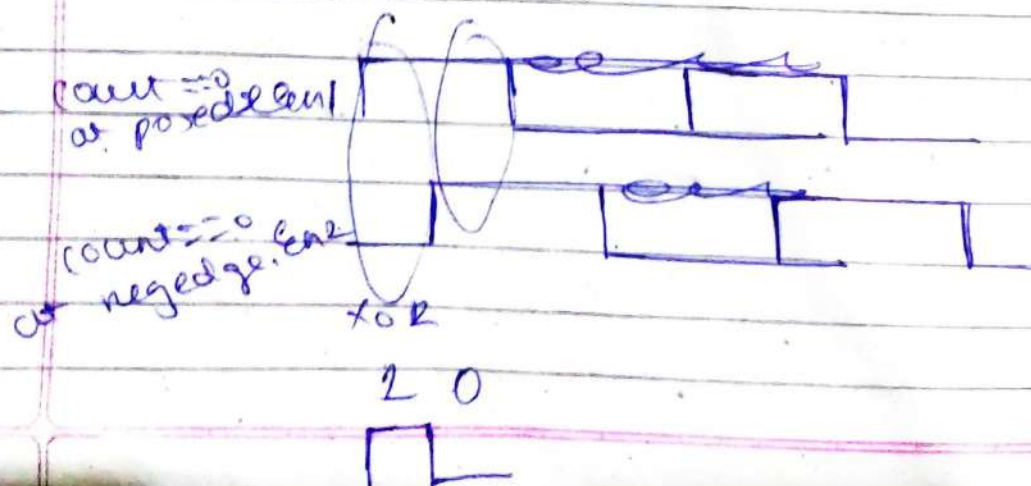
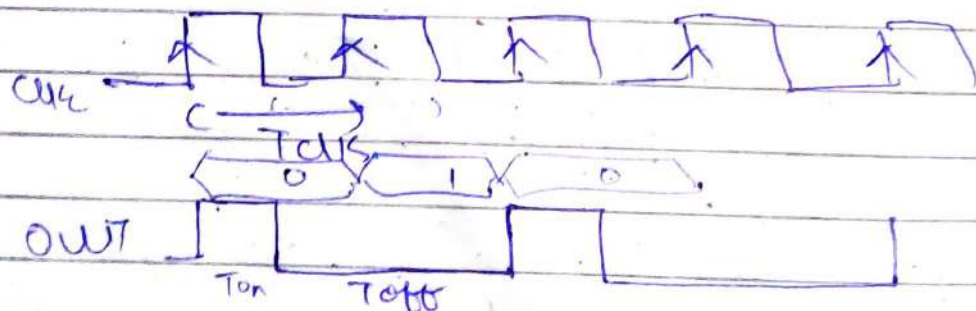
$$T_{out} = 2 T_{clk}$$

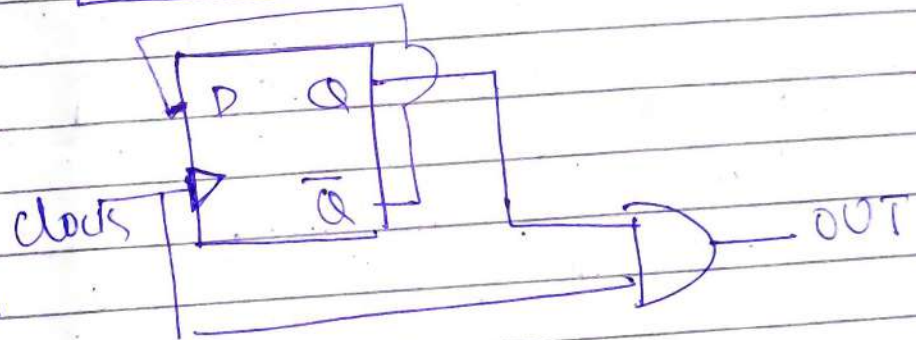
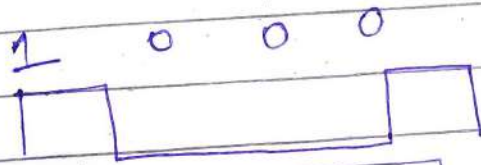
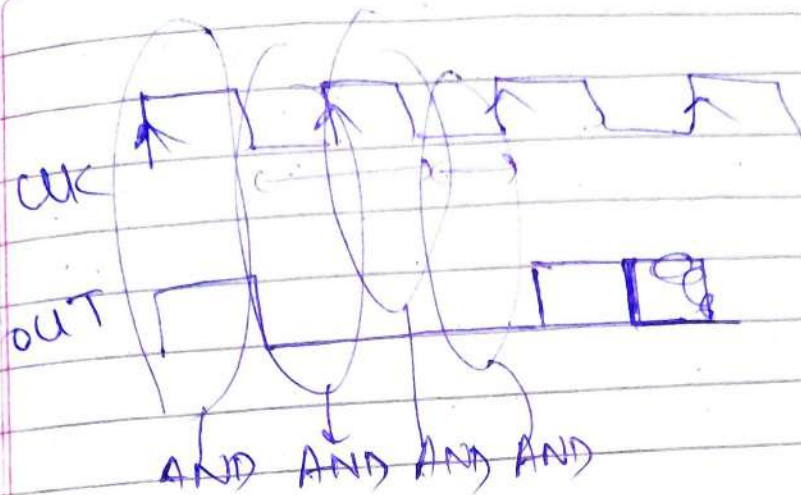
$$= \frac{T}{4}$$

$$T_{on} = \frac{(2 T_{clk})}{4/2} = \frac{T_{clk}}{2}$$

$$T_{off} = T - \frac{T}{4} = \frac{3T}{4}$$

$$= 3 \left(\frac{2 T_{clk}}{4} \right) = \frac{3 T_{clk}}{2}$$





$Q=0 \rightarrow D=1 \rightarrow Q=1$
 $Q=1 \rightarrow D=1 \rightarrow Q=1$

§ $f/3$ with $\frac{1}{3} DT$

$$f_{out} = \frac{f}{3}$$

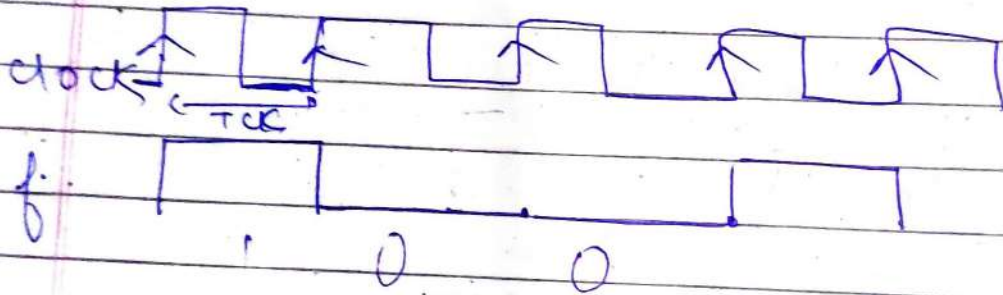
$$DT = \frac{1}{3} = \frac{T_{on}}{T}$$

$$T = 3 T_{clk}$$

$$T_{on} = \frac{1}{3} T$$

$$= \frac{1}{3} (3 T_{clk})$$

$$= 1 T_{clk}$$



How to Design Hardware??

Synchronous if we observe

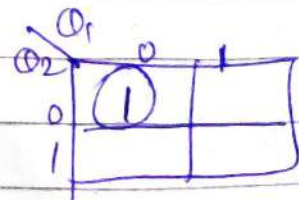
• we can use FSM Design

$a, b, c \rightarrow$ States.

$00, 01, 10 \rightarrow$ state representation

PS NS O/P $\rightarrow$ State Table.

$q_2 q_1$	$q_2^+ q_1^+$	Output
00	01	1
01	10	0
10	00	0

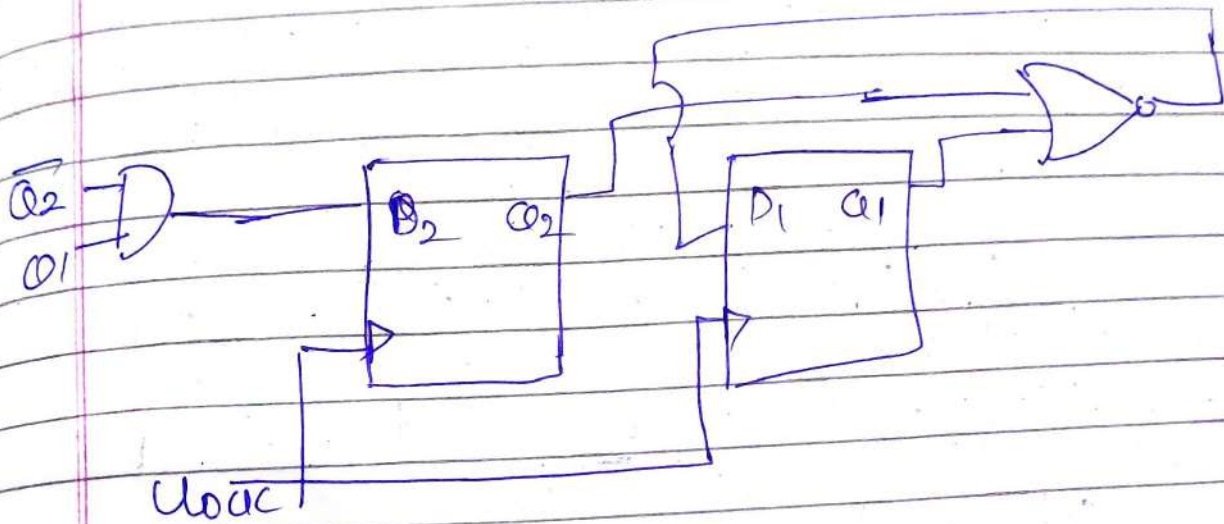


$$y = \overline{q_2} \overline{q_1} = \overline{q_2} + \overline{q_1}$$

If there are 3 states then

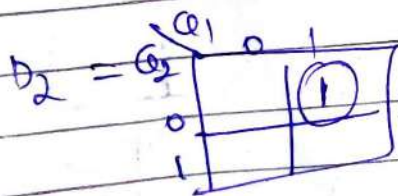
How many f.f required ??

2 flipflops

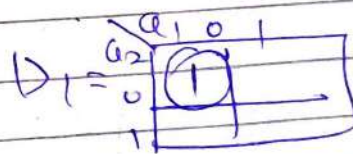


PS	NS	I/P	output	excitation table
$Q_2 Q_1$	$Q_2^+ Q_1^+$	$D_2 D_1$	y	
00	01	0 1	1	
01	10	1 0	0	
10	00	0 0	0	

$$y = Q_2 + Q_1$$



$$D_2 = Q_2 \cdot Q_1$$



$$D_1 = \overline{Q_2} \overline{Q_1} = \overline{Q_2 + Q_1}$$

Q $\frac{f}{30}$ with 50% DT

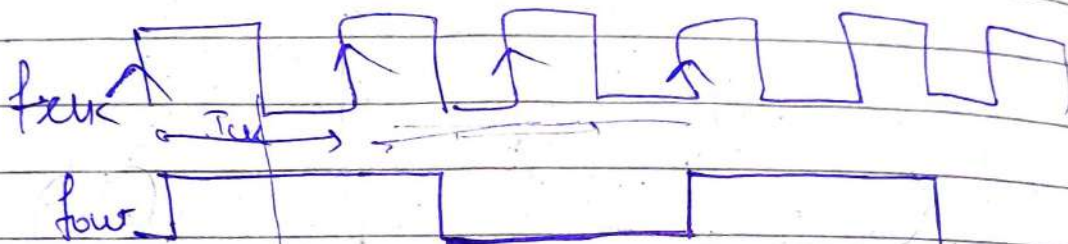
$$f_{out} = \frac{f_{ave}}{3}$$

$$T = 3 T_{clk}$$

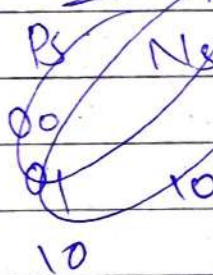
$$T_{orv} = \frac{1}{2} T$$

$$= \left(\frac{3}{2} \right) T_{clk}$$

$$= 1.5$$

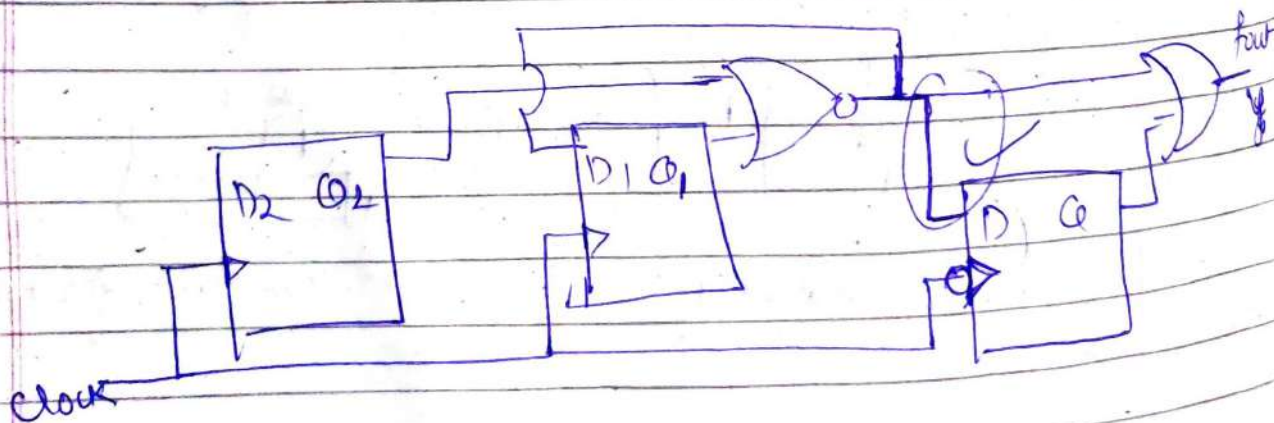


~~f_out~~ 0.33%



Negative edge of 0.33%

this complete signal will pass to another BP which will trigger at negative edge


~~| | P5 | N5 | 10 | 01 |
|----|----|----|----|----|
| D2 | 00 | 01 | 10 | 00 |
| D1 | 00 | 01 | 10 | 00 |
| D | 00 | 01 | 10 | 00 |~~